Problem

To detect ambiguity in Natural-Language Software Requirements using the RoBERTa model

Current Progress

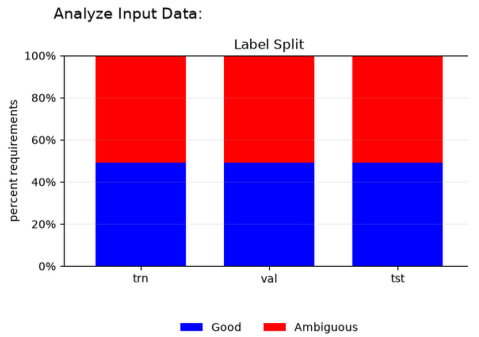
- created a common fine-tuned pipeline for RoBERTa using the annotated PURE dataset, saved the fine-tuned RoBERTa to be used by the group
- analyzed the training dataset (Human / AI-Aided)
- analyzed the test dataset and corresponding results from the fine-tuned RoBERTa

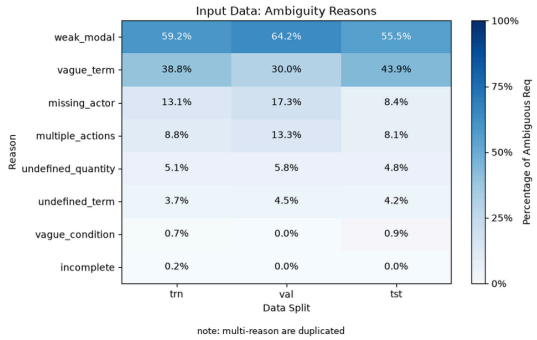
Human Group

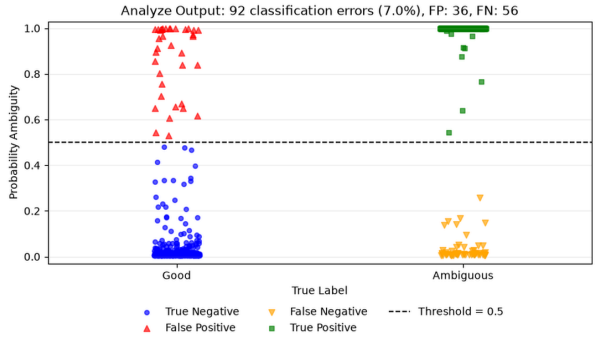
K12148218: Ostap Ohlashennyy	K12434880: Lloyd Aaron Yandoc

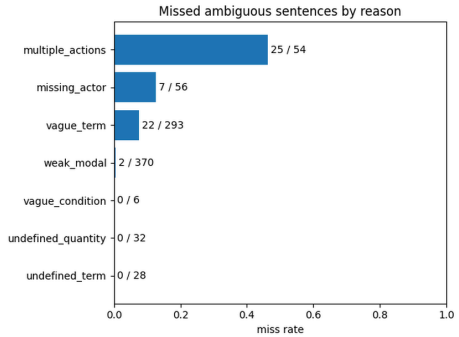
Key Findings

- The label mix is generally balanced across the training, validation, and test sets.
- `weak_modal` is the most frequent annotated ambiguity reason.
- Model probabilities are often near 0 or 1, including for some incorrect predictions.
- `multiple_actions` has the highest observed miss rate.









AI-Aided Group

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Key Findings

- The model missed 25/28 requirements annotated with multiple actions alone, versus 0/26 with additional reasons. The contrast persists within shared-source comparisons and across length bands.
- Clarifications noticeably lowered ambiguity scores for two of four requirements, while paraphrases barely affected them. Only the broader size/timing rewrite changed the predicted class. This small pilot shows wording sensitivity, not reliable improvement.

